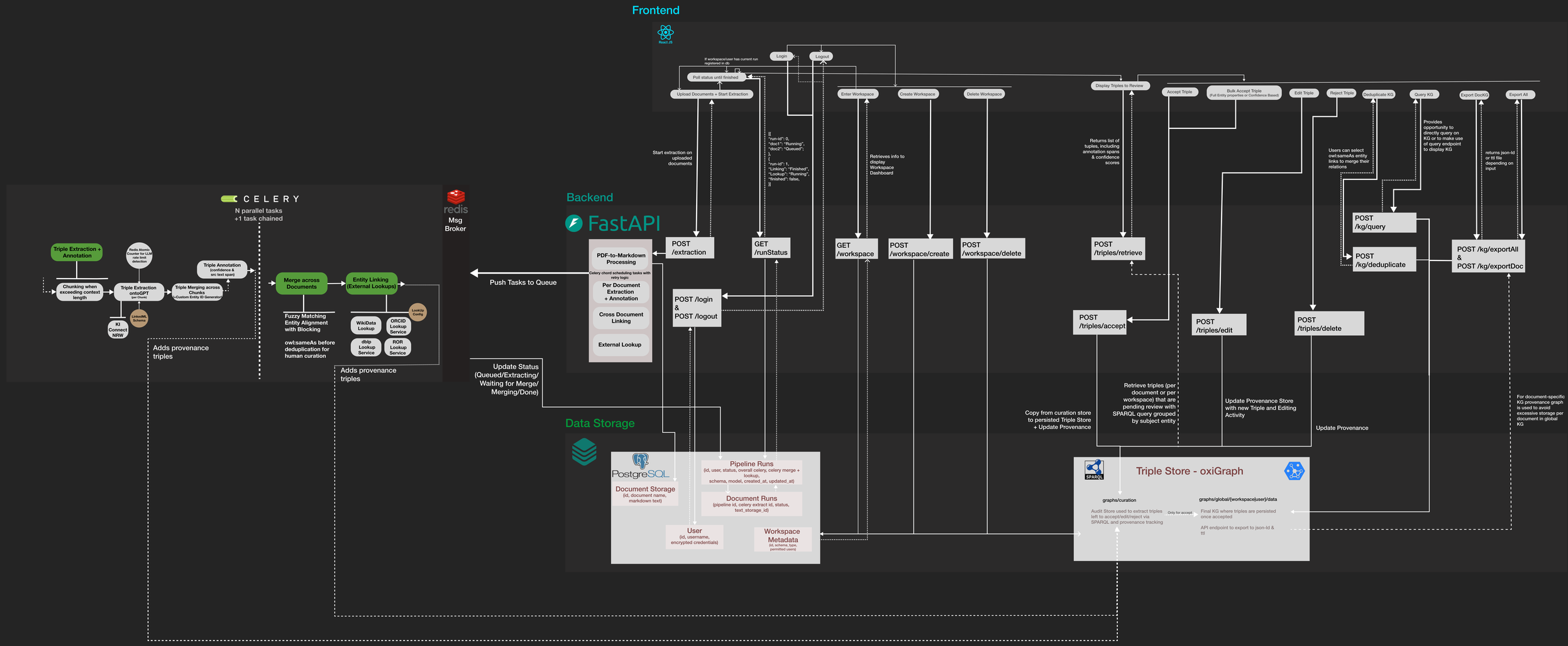
Frontend
React JS
If workspace/user has current run registered in db
Login
Logout
Poll status until finished
Upload Documents + Start Extraction
Enter Workspace
Create Workspace
Delete Workspace
Display Triples to Review
Accept Triple
Bulk Accept Triple
(Full Entity properties or Confidence Based)
Edit Triple
Reject Triple
Deduplicate KG
Query KG
Export DocKG
Export All
Start extraction on uploaded documents
[{ "run-id": 0, "doc1": "Running", "doc2": "Queued"; }, { "run-id": 1, "Linking": "Finished", "Lookup": "Running", "finished": false, }]
Retrieves info to display Workspace Dashboard
Returns list of tuples, including annotation spans & confidence scores
Users can select owl:sameAs entity links to merge their relations
Provides opportunity to directly query on KG or to make use of query endpoint to display KG
returns json-ld or ttl file depending on input
CELERY
N parallel tasks
+1 task chained
redis
Msg Broker
Backend
FastAPI
Triple Extraction + Annotation
Redis Atomic Counter for LLM rate-limit detection
Chunking when exceeding context length
Triple Extraction ontoGPT (per Chunk)
Triple Annotation (confidence & src text span)
Triple Merging across Chunks +Custom Entity ID Generator
KI Connect NRW
LinkonML Schema
Merge across Documents
Fuzzy Matching Entity Alignment with Blocking
owl:sameAs before deduplication for human curation
Entity Linking (External Lookups)
LookUp Config
WikiData Lookup
ORCID Lookup Service
dblp Lookup Service
ROR Lookup Service
Adds provenance triples
Adds provenance triples
Push Tasks to Queue
PDF-to-Markdown Processing
Celery chord scheduling tasks with retry logic
Per Document Extraction + Annotation
Cross Document Linking
External Lookup
POST /extraction
GET /runStatus
POST /login & POST /logout
GET /workspace
POST /workspace/create
POST /workspace/delete
POST /triples/retrieve
POST /kg/query
POST /kg/deduplicate
POST /kg/exportAll & POST /kg/exportDoc
POST /triples/accept
POST /triples/edit
POST /triples/delete
Update Status (Queued/Extracting/ Waiting for Merge/ Merging/Done)
Copy from curation store to persisted Triple Store + Update Provenance
Retrieve triples (per document or per workspace) that are pending review with SPARQL query grouped by subject entity
Update Provenance Store with new Triple and Editing Activity
Update Provenance
For document-specific KG provenance graph is used to avoid excessive storage per document in global KG
Data Storage
PostgreSQL
Document Storage
(id, document name, markdown text)
Pipeline Runs
(id, user, status, overall celery, celery merge + lookup, schema, model, created_at, updated_at)
Document Runs
(pipeline id, celery extract id, status, text_storage_id)
User
(id, username, encrypted credentials)
Workspace Metadata
(id, schema_type, permitted users)
SPARQL
Triple Store - oxiGraph
graphs/curation
Audit Store used to extract triples left to accept/edit/reject via SPARQL and provenance tracking
Only for accept
graphs/global/{workspace|user}/data
Final KG where triples are persisted once accepted
API endpoint to export to json-ld & ttl